

Q.1)
Soln:-

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) [V_{GS} - V_T]^2$$

$$I_{D.} = \frac{1}{2} \times 2 \times 10^{-3}$$

$$= 1 \text{ mA}$$

Q.2)

$$V_0 = V_{DD} - I_D R_D$$

$$g_m = K [V_{GS} - V_T]$$

$$= 2000(1) \times 10^{-6}$$

$$= 2 \times 10^{-3}$$

$$g_m = 2 \text{ mS}$$

Q.3) $I_D = \frac{1}{2} \times 2 \times 10^{-3} [4 - 1]^2$

$$I_D = 9 \text{ mA}$$

Q.4) $= 2000 \times 10^{-6} [3]$

$$= 6 \text{ mS}$$

Q.5)
Soln:-

$$\frac{1}{2} \times 100 \times 10^{-6} \times 4 [5 - V_0 - 1]^2 = \frac{1}{2} \times 100 \times 10^{-6} [V_0 - 0 - 1]^2$$

$$4 = \frac{[V_0 - 1]^2}{[5 - V_0 - 1]^2}$$

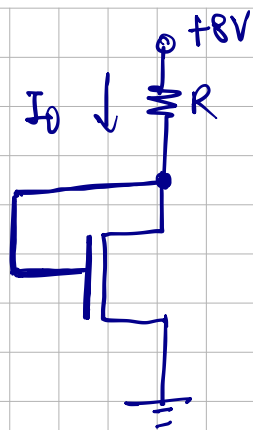
$$V_0 - 1 = 2 [5 - V_0 - 1]$$

$$V_0 - 1 = 8 - 2V_0$$

$$3V_0 = 9$$

$$\boxed{V_0 = 3}$$

Q.6)
Soln:-



$$V_{GS} = 8 - 3 = 5V$$

$$3 \times 10^{-3} = \frac{1}{2} k [4]^2$$

$$\frac{3}{8} \times 10^{-3} = k \quad \text{--- (1)}$$

$$V_D = V_{GS}$$

$$\text{now } V_D = 8 - I_D \times 3k\Omega = V_{GS}$$

$$I_D = \frac{8 - V_{GS}}{3k\Omega}$$

$$\frac{8 - V_{GS}}{3k\Omega} = \frac{1}{2} k [V_{GS} - 1]^2$$

$$\frac{8 - V_{GS}}{3k\Omega} = \frac{1}{2} \times \frac{3}{8} \times 10^{-3} [V_{GS} - 1]^2$$

$$\frac{8 - V_{GS}}{3} = \frac{3}{16} [V_{GS} - 1]^2$$

$$\frac{16}{9} [8 - V_{GS}] = V_{GS}^2 + 1 - 2V_{GS}$$

$$\frac{16 \times 8}{9} - \frac{16}{9} V_{GS} + 2V_{GS} = V_{GS}^2 + 1$$

$$V_{GS}^2 + \frac{16}{9} V_{GS} - 2V_{GS} + 1 - \frac{16 \times 8}{9} = 0$$

$$V_{GS}^2 - \frac{2}{9} V_{GS} - \frac{119}{9} = 0$$

$$9 V_{GS}^2 - 2 V_{GS} - 119 = 0$$

using formula for v_{DSAT}

$$V_D \approx 3.6V$$

$$\text{now } I_D = \frac{8 - 3.6}{3k\Omega} = \frac{4.4}{3k}$$

$$I_D = 1.47 \text{ mA.}$$

Q.7]

Soln:- It's a current mirror circuit, as M_1 & M_2 are identical with same Gate voltage,

So same current will flow through the circuit.

$$\therefore I_1 = I_{\text{bias}}$$

Q.8]